Calculation of genotype values (BLUEs) emmeans a pisum

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Setup

Packages

```
suppressPackageStartupMessages(library("tidyverse"))
suppressPackageStartupMessages(library("lme4"))
suppressPackageStartupMessages(library("Hmisc"))
suppressPackageStartupMessages(library("corrplot"))
suppressPackageStartupMessages(library("emmeans"))
suppressPackageStartupMessages(library("performance"))
suppressPackageStartupMessages(library("MASS"))
```

More than 4 reps

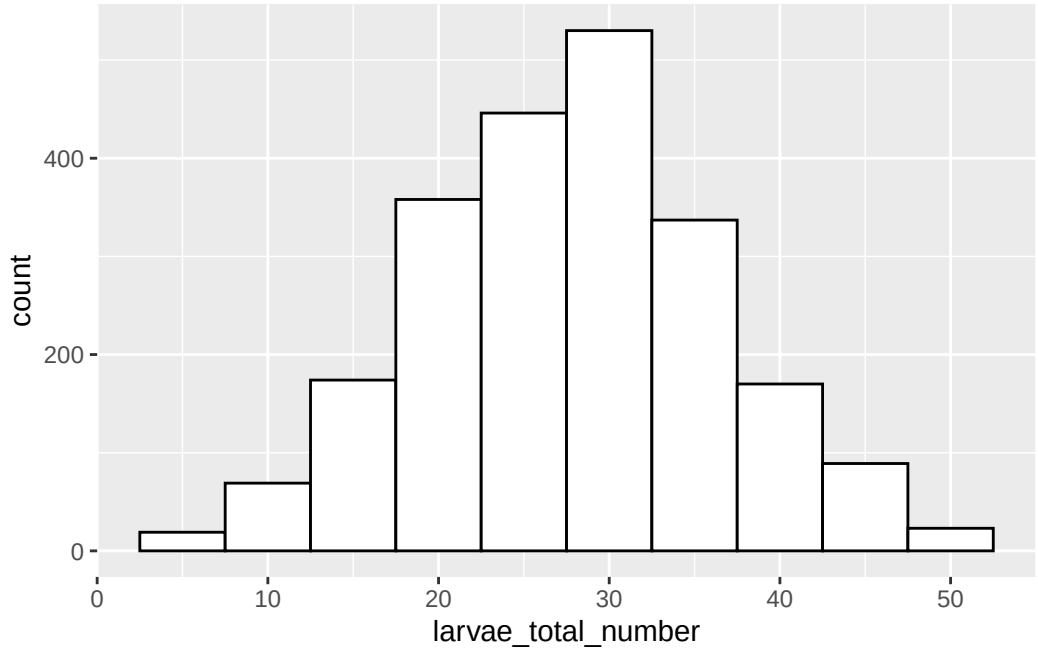
Filtered data import

```
data250pisum_4 <- read.csv("../../../01_phenotipic_analysis/01.dataset_filtering/02_A.pisum/
data250pisum_4$genotype <- as.factor(data250pisum_4$ivip)

data250pisum_4 <- data250pisum_4 %>%
  mutate(ivip = ifelse(seed_batch == "KF2229", "hedin", ivip))
```

Data distribution

```
data250pisum_4 %>%
  ggplot(., aes(x = larvae_total_number)) +
  geom_histogram(binwidth = 5, fill = "white", color = "black")
```



Linear model fitting and emmeans extraction

Useful lesson on LMMs.

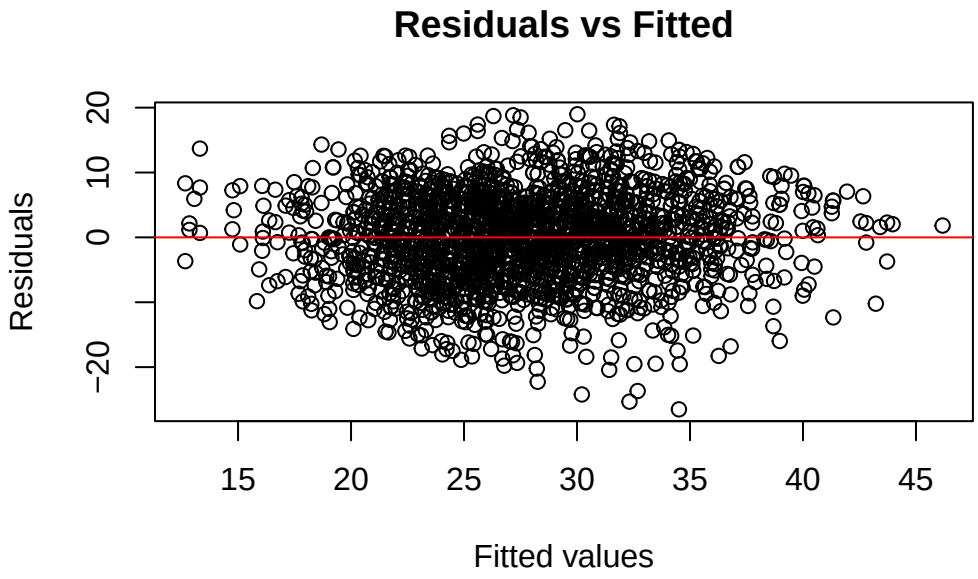
Fit linear models

We tested few model to see which one is better to estimate the larvae fecundity

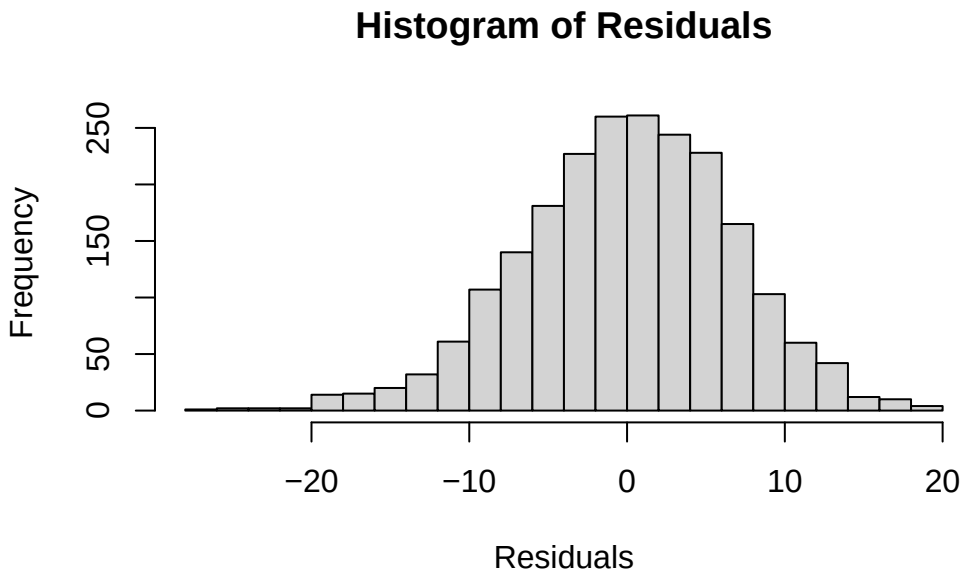
```
model <- lmer(larvae_total_number ~ ivip + (1 | session/block), data = data250pisum_4)
```

```
# Extract fitted values and residuals
fitted_values <- fitted(model)
residuals_values <- residuals(model)

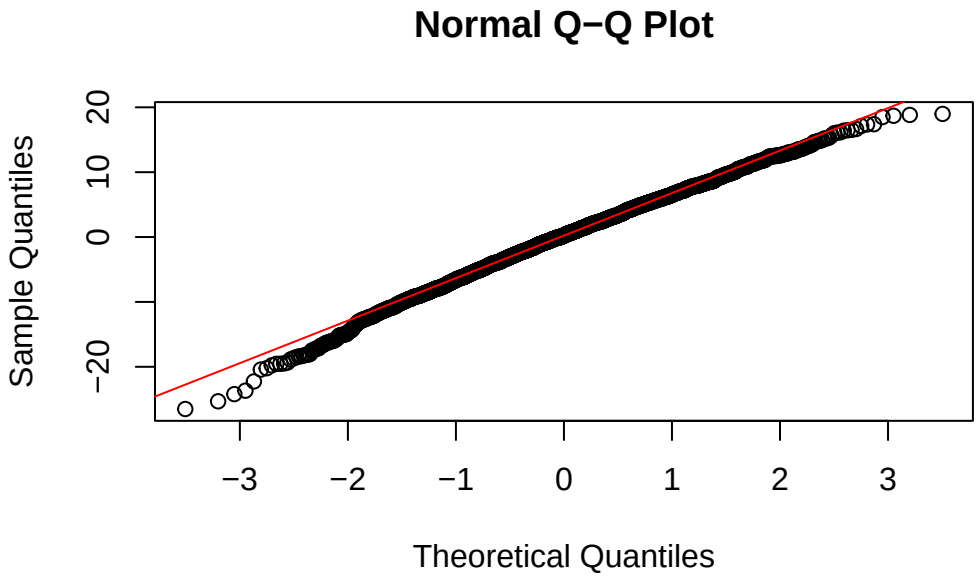
#homosedasticity
# Plot residuals vs fitted values
plot(fitted_values, residuals_values,
     main="Residuals vs Fitted",
     xlab="Fitted values",
     ylab="Residuals")
abline(h=0, col="red")
```



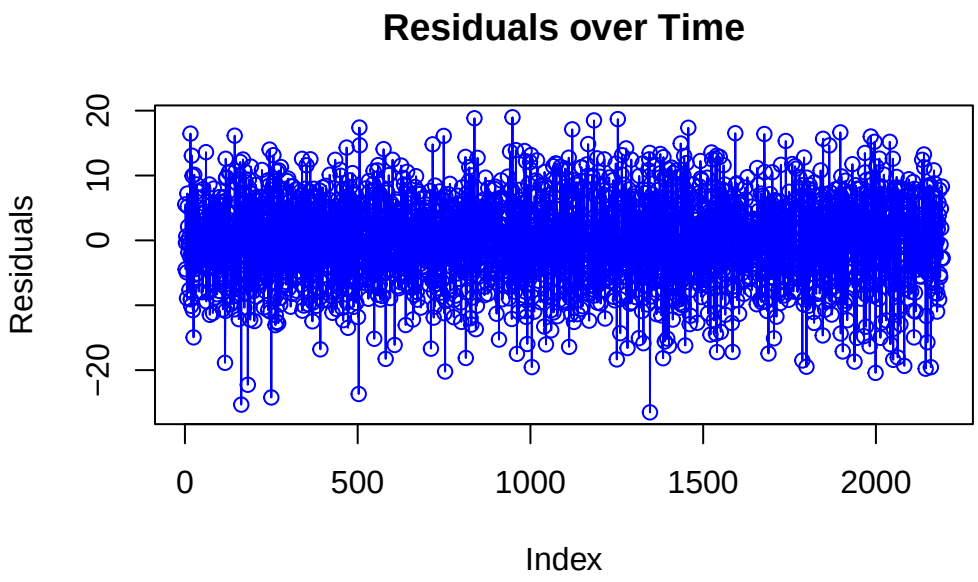

```r
#residual normality
hist(residuals_values, main="Histogram of Residuals", xlab="Residuals", breaks=20)
```



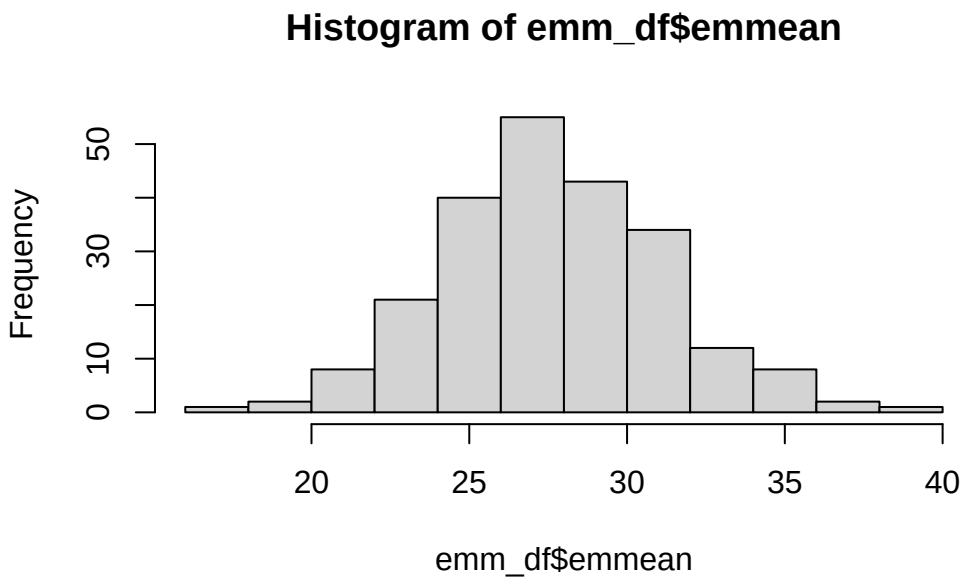
```r
# 2. Q-Q Plot for Residuals
qqnorm(residuals_values)
qqline(residuals_values, col = "red")
```




```r
#residual independance
# Plot residuals over time (for time-series data)
plot(residuals(model), type="o", col="blue", main="Residuals over Time", xlab="Index", ylab="
```




```
# Extract EMMeans for genotype
emm_genotype <- emmeans(model, ~ ivip)
emm_df <- as.data.frame(emm_genotype)
hist(emm_df$emmean)
```




```
emm_df <- emm_df[,c(1,2)]

# Define the file path
file_path <- "~/Documents/gitrepo/faba_specifics/02.analyses/01.GWAS/01_phenotipic_analysis/

# Save the data frame to the specified location
  write.csv(emm_df, file = file_path, row.names = FALSE)

  genotype_name<- emm_df[,1]

file_path <- "~/Documents/gitrepo/faba_specifics/02.analyses/01.GWAS/01_phenotipic_analysis/

write.csv(emm_df, file = file_path, row.names = FALSE)
```

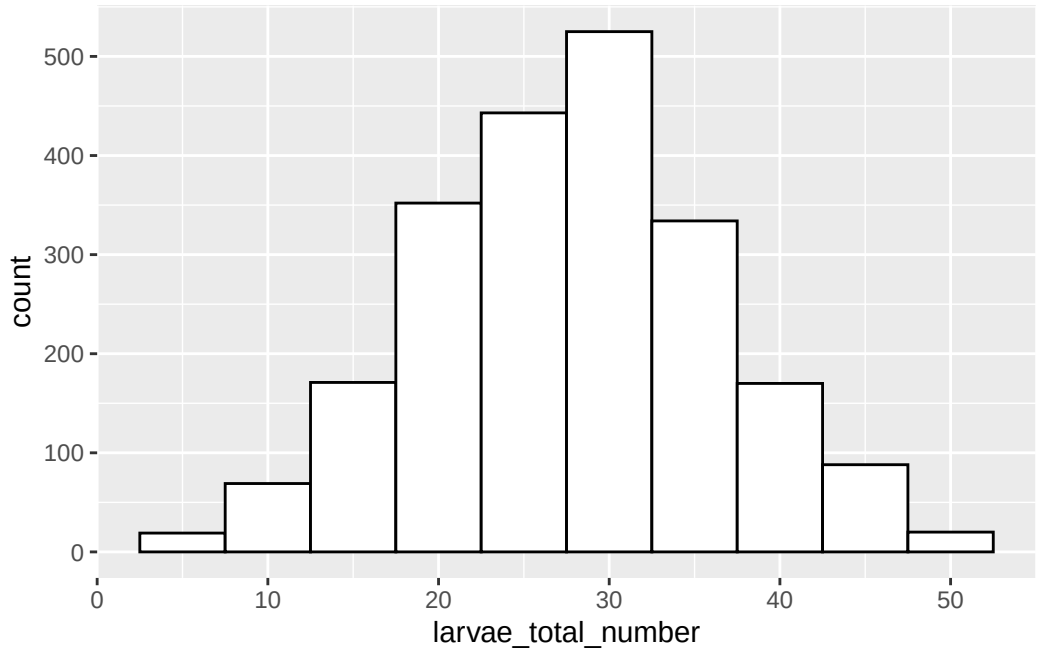
More than 5 reps

Filtered data import

```
data250pisum_5 <- read.csv("../../../01_phenotipic_analysis/01.dataset_filtering/02_A.pisum/
data250pisum_5$genotype <- as.factor(data250pisum_5$ivip)
```

Data distribution

```
data250pisum_5 %>%
  ggplot(., aes(x = larvae_total_number)) +
  geom_histogram(binwidth = 5, fill = "white", color = "black")
```



Linear model fitting and emmeans extraction

Useful lesson on LMMs.

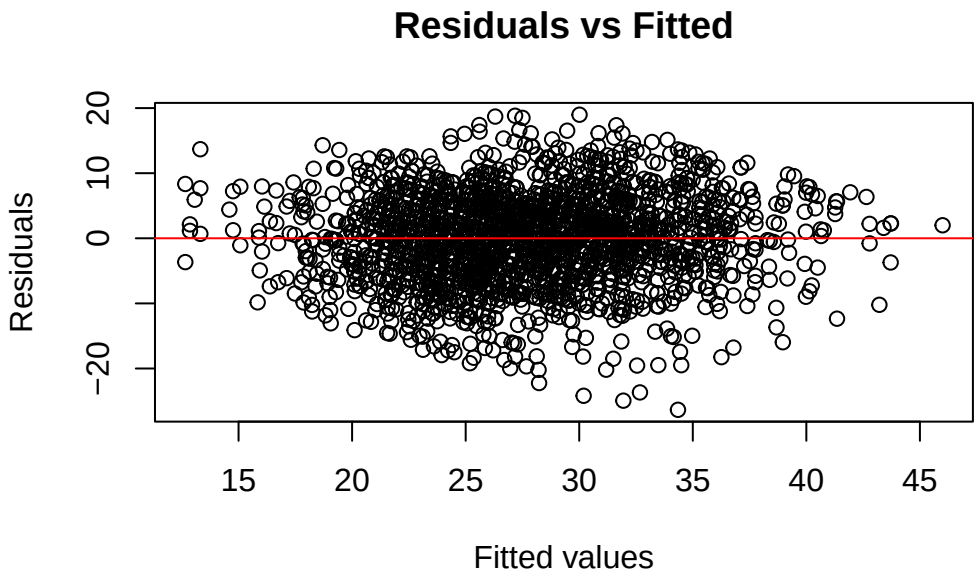
Fit linear models

We tested few model to see which one is better to estimate the larvae fecundity

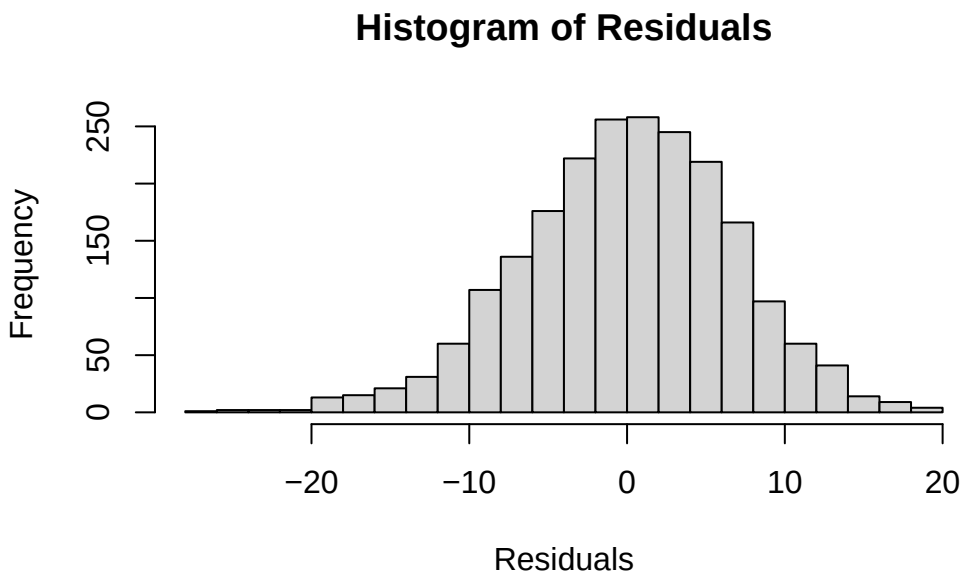
```
model <- lmer(larvae_total_number ~ ivip + (1 | session/block), data = data250pisum_5)
```

```
# Extract fitted values and residuals
fitted_values <- fitted(model)
residuals_values <- residuals(model)
```

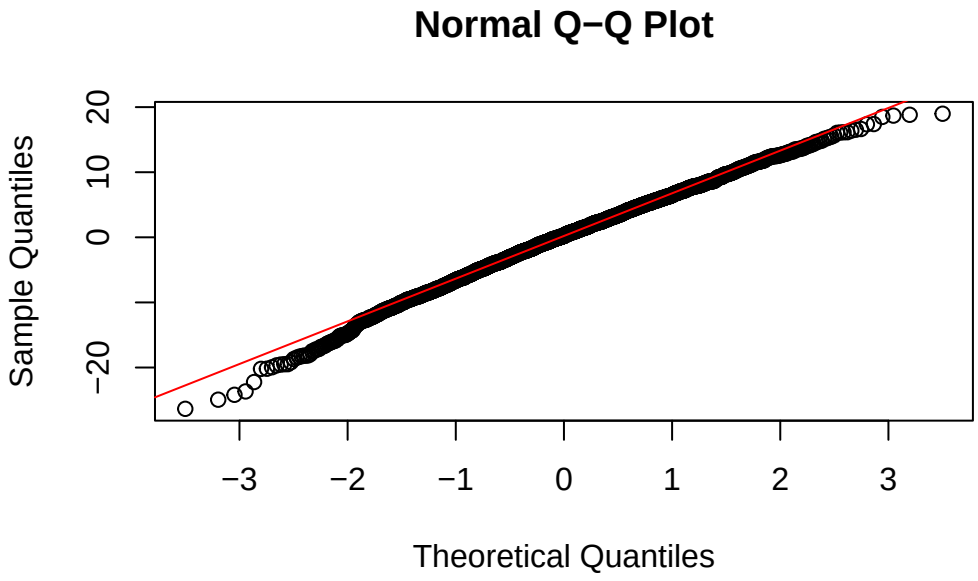
```r
#homosedasticity
# Plot residuals vs fitted values
plot(fitted_values, residuals_values,
     main="Residuals vs Fitted",
     xlab="Fitted values",
     ylab="Residuals")
abline(h=0, col="red")
```



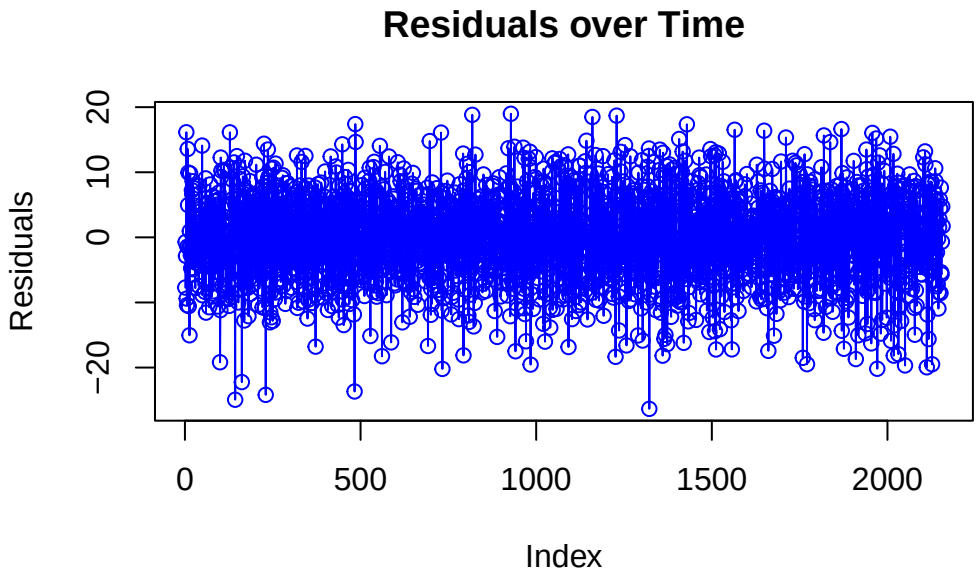

```r
#residual normality
hist(residuals_values, main="Histogram of Residuals", xlab="Residuals", breaks=20)
```



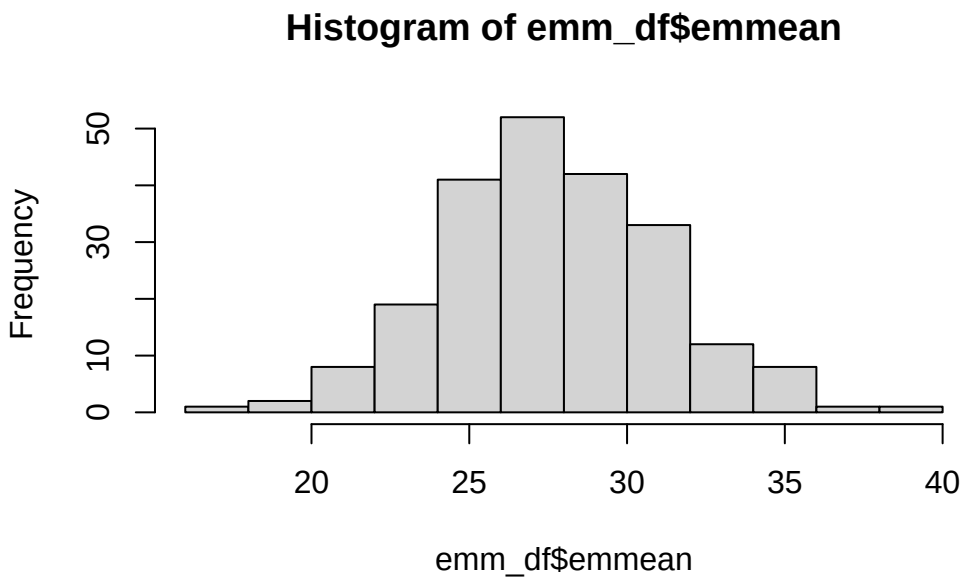

```r
# 2. Q-Q Plot for Residuals
qqnorm(residuals_values)
qqline(residuals_values, col = "red")
```



```r
#residual independance
# Plot residuals over time (for time-series data)
plot(residuals(model), type="o", col="blue", main="Residuals over Time", xlab="Index", ylab="
```




```r
# Extract EMMeans for genotype
emm_genotype <- emmeans(model, ~ ivip)
emm_df <- as.data.frame(emm_genotype)
hist(emm_df$emmean)
```




```
emm_df <- emm_df[,c(1,2)]

# Define the file path
file_path <- "~/Documents/gitrepo/faba_specifics/02.analyses/01.GWAS/01_phenotipic_analysis/

# Save the data frame to the specified location
write.csv(emm_df, file = file_path, row.names = FALSE)

  genotype_name<- emm_df[,1]

file_path <- "~/Documents/gitrepo/faba_specifics/02.analyses/01.GWAS/01_phenotipic_analysis/

write.csv(emm_df, file = file_path, row.names = FALSE)
```

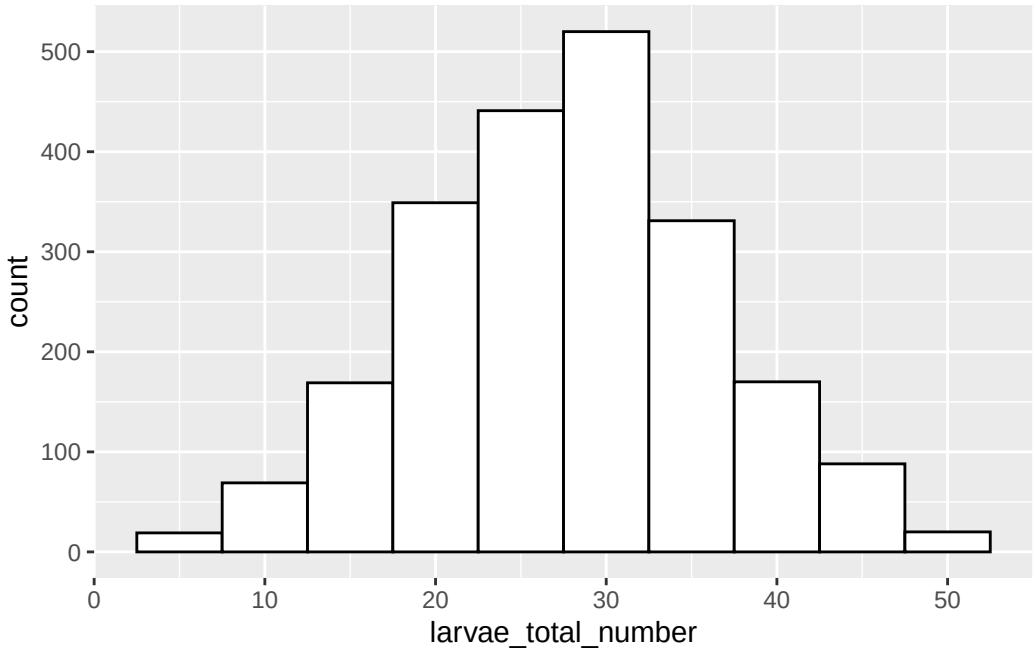
More than 6 reps

Filtered data import

```
data250pisum_6 <- read.csv("../../../01_phenotipic_analysis/01.dataset_filtering/02_A.pisum/
data250pisum_6$genotype <- as.factor(data250pisum_6$ivip)
```

Data distribution

```
data250pisum_6 %>%
  ggplot(., aes(x = larvae_total_number)) +
  geom_histogram(binwidth = 5, fill = "white", color = "black")
```



Linear model fitting and emmeans extraction

Useful lesson on LMMs.

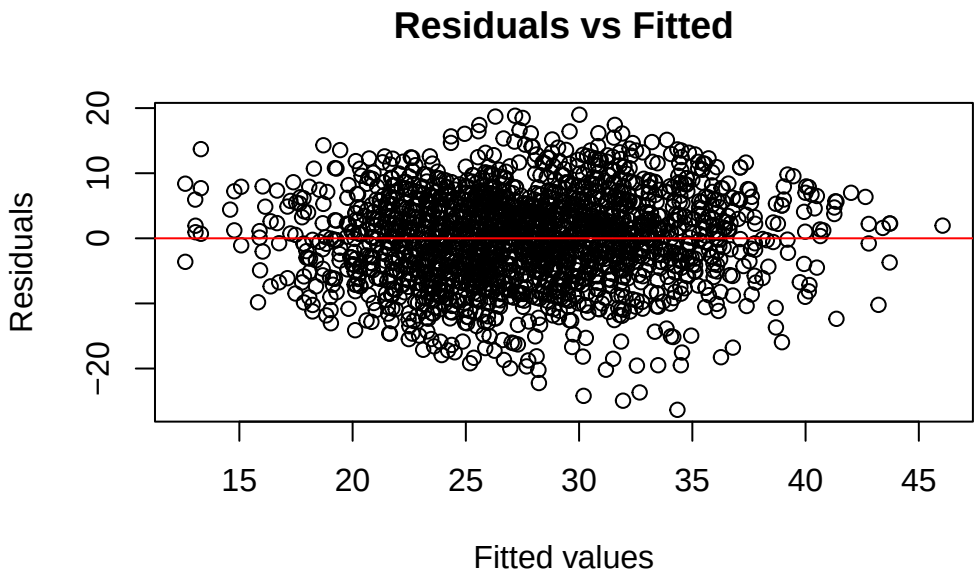
Fit linear models

We tested few model to see which one is better to estimate the larvae fecundity

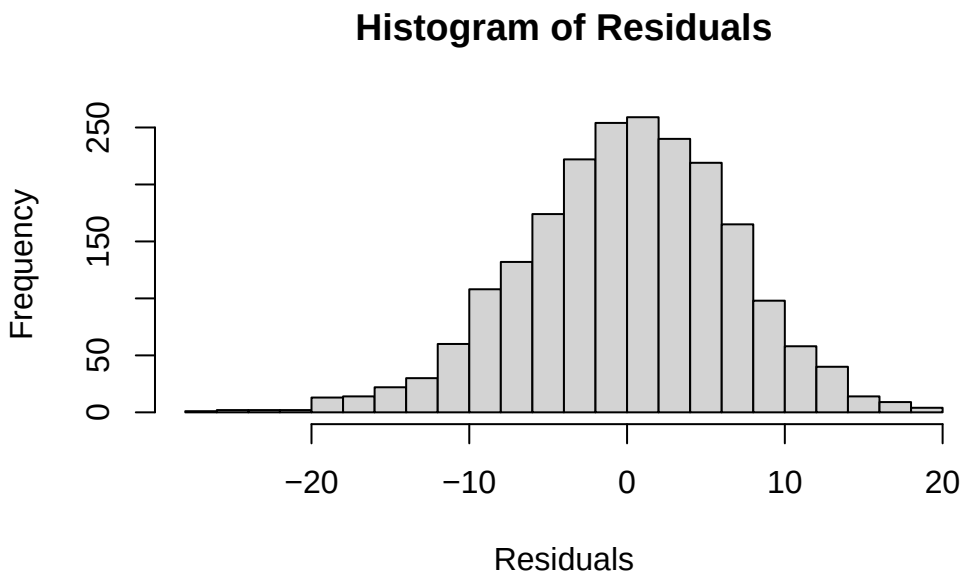
```
model <- lmer(larvae_total_number ~ ivip + (1 | session/block), data = data250pisum_6)
```

```
# Extract fitted values and residuals
fitted_values <- fitted(model)
residuals_values <- residuals(model)
```

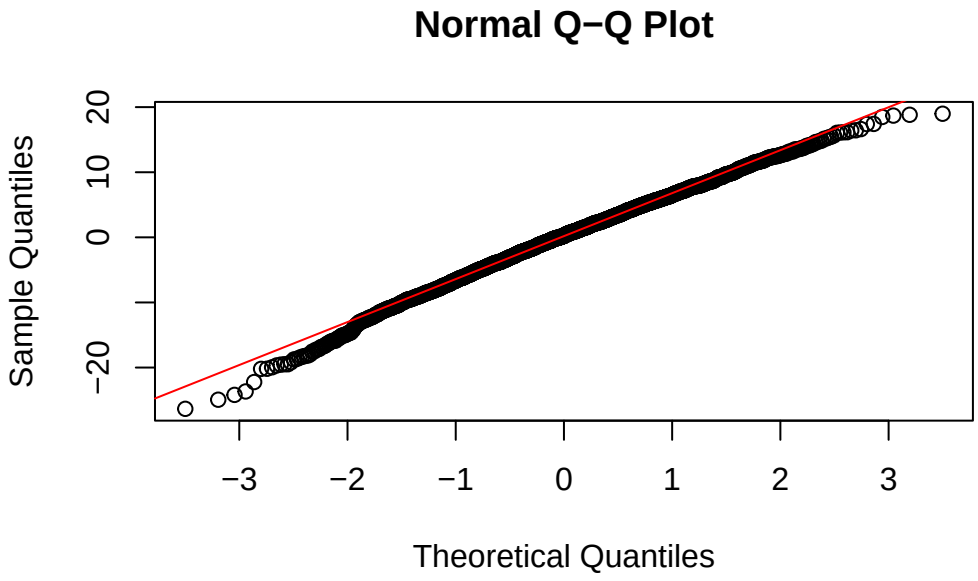
```r
#homosedasticity
# Plot residuals vs fitted values
plot(fitted_values, residuals_values,
     main="Residuals vs Fitted",
     xlab="Fitted values",
     ylab="Residuals")
abline(h=0, col="red")
```



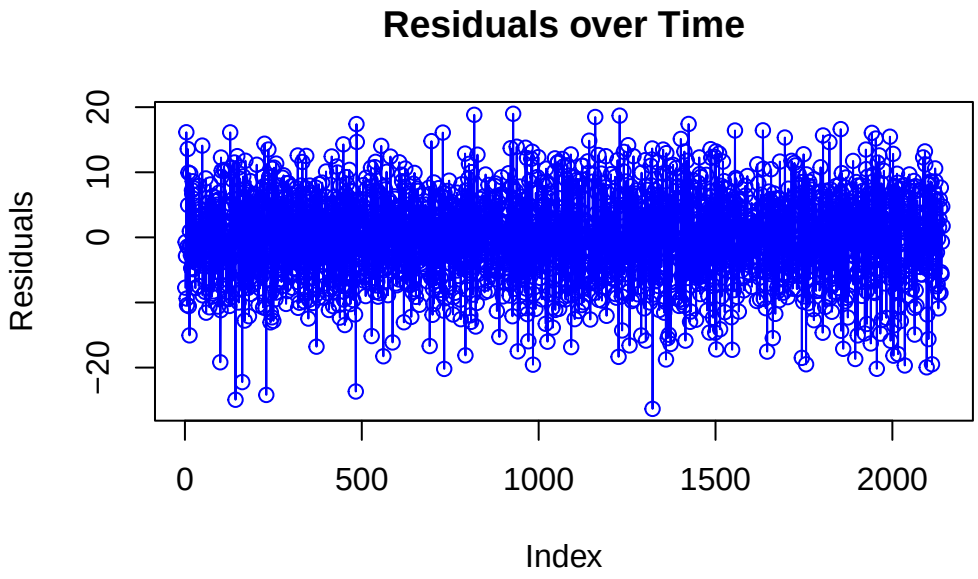

```r
#residual normality
hist(residuals_values, main="Histogram of Residuals", xlab="Residuals", breaks=20)
```



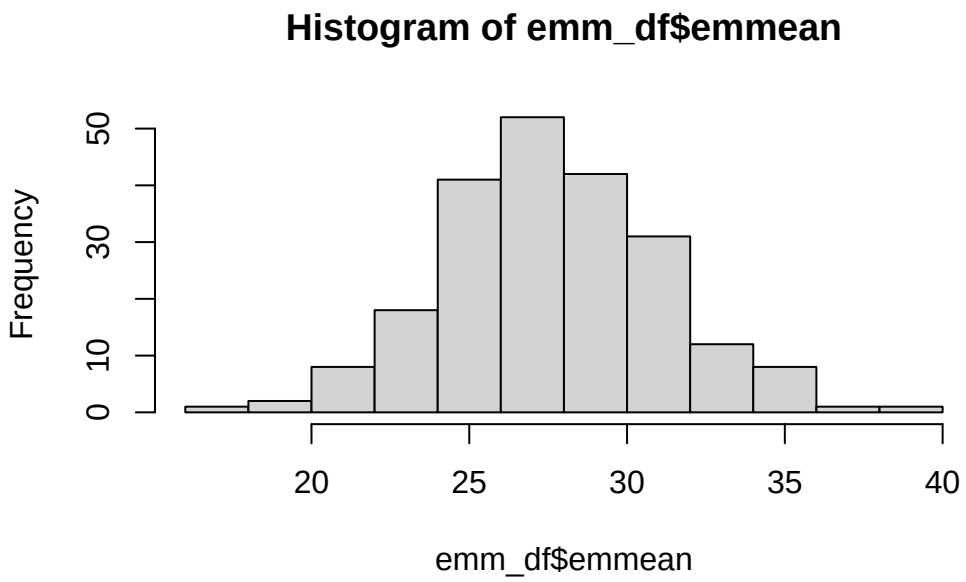

```r
# 2. Q-Q Plot for Residuals
qqnorm(residuals_values)
qqline(residuals_values, col = "red")
```



```r
#residual independance
# Plot residuals over time (for time-series data)
plot(residuals(model), type="o", col="blue", main="Residuals over Time", xlab="Index", ylab="
```




```r
# Extract EMMeans for genotype
emm_genotype <- emmeans(model, ~ ivip)
emm_df <- as.data.frame(emm_genotype)
hist(emm_df$emmean)
```




```r
emm_df <- emm_df[,c(1,2)]

# Define the file path
file_path <- "~/Documents/gitrepo/faba_specifics/02.analyses/01.GWAS/01_phenotipic_analysis/

# Save the data frame to the specified location
write.csv(emm_df, file = file_path, row.names = FALSE)

  genotype_name<- emm_df[,1]

file_path <- "~/Documents/gitrepo/faba_specifics/02.analyses/01.GWAS/01_phenotipic_analysis/

write.csv(emm_df, file = file_path, row.names = FALSE)
```